



Automatic liver and tumour segmentation from CT images using Deep learning algorithm

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ARTICLE INFO

Keywords:

Liver & tumour segmentation
CNN
Deep learning
CT image
ResUNet
DSC

ABSTRACT

The diagnosis and treatment of liver diseases from computed tomography (CT) images is an indispensable task for segmentation of Liver & its tumours. Due to the uneven presence, fuzzy borders, diverse densities, shapes and sizes of lesions segmentation of liver & its tumour is a difficult task. At this point we mainly focused on deep learning algorithms for segmenting liver and its tumour from abdominal CT scan images thereafter minimising the time & energy used for a liver diseases diagnosis. The algorithm is used here is based on the modified ResUNet architecture. Here we present, an automatic method based on semantic segmentation convolutional neural networks (CNNs) to segment Liver from CT scans and lesions from segmented liver part. The proposed system attains a Dice Similarity Coefficients (DSCs) of 96.35% and 89.28% and accuracy of 99.71% and 99.72% for liver and tumour segmentations, respectively. Comparison with the linked methods confirms the promise of the proposed system for liver and tumour segmentations.

1. Introduction

According to the WHO information, liver cancer has been the second most important cause of death in all cancers [1]. An accurate and robust tumours segmentation method is needed for effective diagnosis and treatment. Consequently, premature analysis and precise review of tumours are critical. But, due to the minor noticeable deviations among healthy tissues and tumoral ones the detection of liver and its tumours is challenging. To detect the tumours a good and repeatable technique would be an advantage. In order to evade various and consecutive segmentations a clinically motivated approach must work for various tumour types which is a quite difficult task at the same time.

First, Liver and its various tumours have to be segmented, with varied appearances. Then, often a striking resemblance exists between tumoral and healthy tissues, however, some tumours are only visible, or more distinct, when a suitable phase has been considered. Another exciting problem is due to the liver itself. Being an organ with a high grade of vascularity, lack of visible edges on most borders, the images obtained are really noisy. At last, because of the focus on medical use, the technique should work for images in clinical routine, meaning images from diverse CT machines, with various levels of resolution and for both overlapping as well as non-overlapping slices. The segmentation of liver & diverse tumour types will be achieved while satisfying clinical constraints. In order to satisfy the speed of physicians the entire segmentation process should not take more than few minutes. However, in present scenario, segmentation is mainly finished manually, which is time-consuming and expensive [2].

In this paper an automatic segmentation method using deep learning is proposed which segments the liver and its tumours using the modified ResUNet model. In addition to the precise automated tumour segmentation, the projected system shows an ability to accurately segment the liver, evidenced by the highest achieved Dice Similarity Coefficient (DSC) metric, compared to the related

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work for tumour and liver segmentation. The rest of this paper is organised as follows. Section 2 overviews the literature survey. Section 3 illustrates the methodology. Section 4 details the experimentations, the comparative results. Finally, Section 5 concludes the paper.

2. Literature survey

This segment summaries the related traditional and deep learning techniques for liver and tumour segmentation.

Performing feature extraction is one of the traditional techniques for liver and tumour segmentation, followed by classification. To segment the liver Bastian et al. [3] used intensity feature later applied a super-pixel Simple Linear Iterative Clustering (SLIC) approach and an AdaBoost algorithm to achieve a Dice rate of 92.13% on 16 abdomen CT test images. Ali et al. [4] to extract the CT liver boundary used first order statistical features of the liver image, later for lesions classification a k-Mean classifier based on the distance and colour is used. Chang et al. [5] segmented the tumours using a binary logistic regression analysis (a region growing algorithm), based on the extraction of texture, shape, and kinetic curve features, to classify the segmented tumours into benign or malignant. To define the complexity of tumour segmentation problem the extracted features may not be sufficient this is drawback of traditional approaches. Hence, more sophisticated features must be investigated.

Deep learning approaches makes full use of the database, i.e., to train a set of convolutional features that can describe the segmentation problem, the CT images and the corresponding Ground Truth (GT) segmentations by the radiologists. Generally, to learn the problem deep approaches use a huge number of parameters (i.e., millions). In the last decade, Convolutional Neural Networks (CNN) architectures have reached outstanding performance in liver and lesions segmentations.

Liver and lesion segmentation approaches using deep learning is discussed here.

(a) To Separate liver or lesion segmentation approaches Liver segmentation is a primary step in many CAD systems for liver cancer or other liver diseases. Hu et al. [6] worked on a 3D CNN model, to provide an initial prior segmentation to segment the liver. Based on image appearance to provide the final segmentation, surface of the prior segmentation was adaptability evolved. Further, throughout the literature different deep learning procedures for lesion segmentation, were implemented. Bi et al. [7] used a deep residual network (Resnet) [8] for liver lesions segmentation. Han et al. [9] segmented the liver lesions automatically by building a CNN model.

Here we are focusing on joint liver and lesion segmentation, a more intensive outline of the related joint methods is provided below.

(b) Joint approaches for liver and lesion segmentations, these approaches can be categorised as one stage or cascaded. Here we will briefly outline each category. One stage deep learning approach attempt to segment the liver and its lesions using one deep CNN stage. Korabelnikov et al. [10] obtained binary segmented image using a segmentation framework which is consisting of a pre-processing step, a pixel-wise classification using pre-trained CNN (Alexnet) and smoothing and thresholding as post-processing step. Badrinarayanan et al. [11] used Cascaded deep learning approaches attempt to segment the liver and lesion in more than one stage by using VGG Segnet model. Bellver et al. [12] used two cascaded VGG16 models, the first one produces the liver segmentation, & it will be the input for the second VGG16 model, which produces the lesion segmentations. Vorontsov et al. [13] used a two cascaded fully convolutional networks (Unet-like) model for the joint segmentation of the liver and lesions from CT images. The model was built [14] that were trained end-to-end to segment the liver and lesions, respectively. In detail, different traditional and deep learning methods used either for liver segmentation or tumour segmentation or both.

However, there is a need to examine better methods to improve the accuracy and dice coefficient. To achieve this task, the proposed framework uses a CNN network, using modified ResUNet model.

3. Methodology

Original dicom images & nii format images from the dataset converted into both JPG & PNG format and then the images are resized to 256×256 size. Where Liver and tumour segmentation consist of two data folders as shown in the flowchart. The first data folder contains original images (JPG) and second data folder contain ground truth images (PNG). Both folders are trained using Modified residual Unet by adjusting different training options. Then the model is tested using a set of images which are not used for training later evaluating the different performance parameters to assess the segmentation. The proposed framework presented below in Fig. 1.

3.1. Dataset

To assess the success of projected semantic segmentation model, the publicly available data-sets are used for training and testing deep learning. The LiTS (Liver tumour segmentation) [15] data-set commonly available public dataset which comprises of 131 contrast enhanced abdominal CT scans from numerous medical sites round the globe. The contest offers reference annotations for the liver contours as well as for liver lesions. To ease training and testing, 760 images including data augmentation are carefully chosen as training samples and 120 images as testing samples where slice thickness of dataset is 0.7–5 mm.

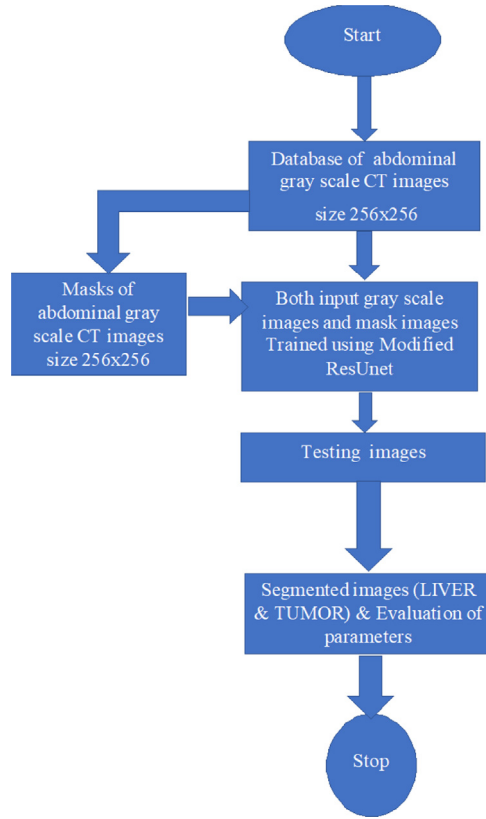


Fig. 1. Methodology flowchart.

3.2. Data arrangements

The algorithm developed has been trained and tested by using images which have been prepared from the available dataset. The three-dimensional images from the LITS dataset contains no distinct mask for liver and its tumour, as a substitute to it under the segmentation file in the dataset they are marked on the same mask. The LITS datasets are converted into 2D png & jpg format from .nii format. The Patient data from the dataset, with no liver and tumour in the mask are removed & also to decrease the class disparity among the background and foreground, images with no liver information is eliminated taken at the beginning and ending of scanning of each patient.

3.3. Image pre-processing

The images in the LITS dataset are having a size of 512×512 measurement initially & due to limited CPU memory using of these images is always challenging. Consequently, all images are resized having a size of 128×128 , 256×256 measurement and also normalisation is applied on all the images having a value between zero & one. Thus, using a MATLAB software all the collected images from the datasets were processed and analysed.

3.4. Algorithm for segmentation

For liver and its tumour segmentation, the algorithm utilises 2D modified ResUNet architectures. To segment liver and its tumours from the abdominal CT scan images these architectures were designed.

3.5. Network architecture

The modified ResUNet architecture represents a hybrid between the traditional U-Net and the ResNet models, where it provides the advantages of both models in one model by replacing convolution blocks with residual blocks. The residual unit will ease the training of CNN and the skip connections among low and high levels of the network and during information propagation inside the residual unit eliminates degradation which leads to less training parameters. The modified ResUNet is constructed by using three paths which includes, encoder which encodes the input to a solid representation, decoder which represents a pixel-wise fashion and

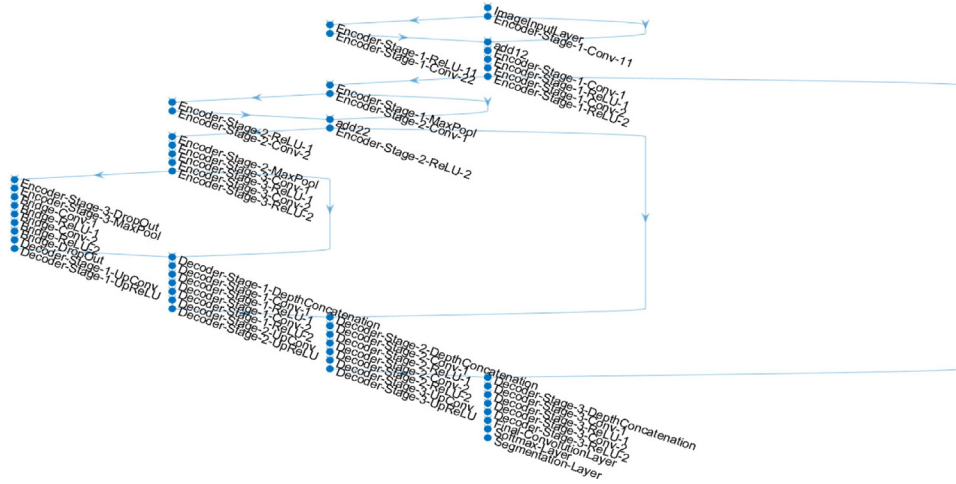


Fig. 2. Schematic representation of the proposed network.

the bridge which connects both encoder and decoder paths together. The schematic representation of the proposed network together as shown in Fig. 2.

4. Results and discussions

4.1. Performance metrics

To assess the performance of segmentation method, a binary mask from the segmented output is equated with ground truth mask to estimate their resemblance. Various metrics like DSC(dice similarity coefficient), accuracy, symmetric volume difference (SVD) and volumetric overlapping error (VOE) are considered to assess the performance.

Dice similarity coefficient (DSC)

DSC is used between two binary masks to measure the amount of overlap. It is the ratio of size of the two segmentations overlap to the two objects total size. It ranges from zero (no overlap) to one (perfect overlap). It signifies the complete performance of the segmentation [16,17]. Following equation is used to calculate dice.

$$Dice(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|}$$

Accuracy

Ratio of correctly segmented samples to the total number samples represents accuracy. Here one represents good segmentation result almost. Following equation is used to calculate accuracy.

$$Accuracy = \frac{TN + TP}{TN + TP + FN + FP}$$

Where TP is a true positive, TN is a true negative, FN is a false negative, and FP is a false positive.

Relative volume difference (RVD)

The portion of change that occurs between the segmented image with the ground truth is RVD. Here zero represents good segmentation result almost. Following equation is used to calculate accuracy.

$$RVD(X, Y) = \frac{|Y| - |X|}{|X|}$$

Volumetric overlapping error (VOE)

Error of overlapped volume between two sets of voxels is VOE. Value of VOE lies between 0 & 1. Here zero indicates perfect segmentation. Following equation is used to calculate VOE.

$$VOE(X, Y) = 1 - \frac{2|X \cap Y|}{|X| + |Y|}$$

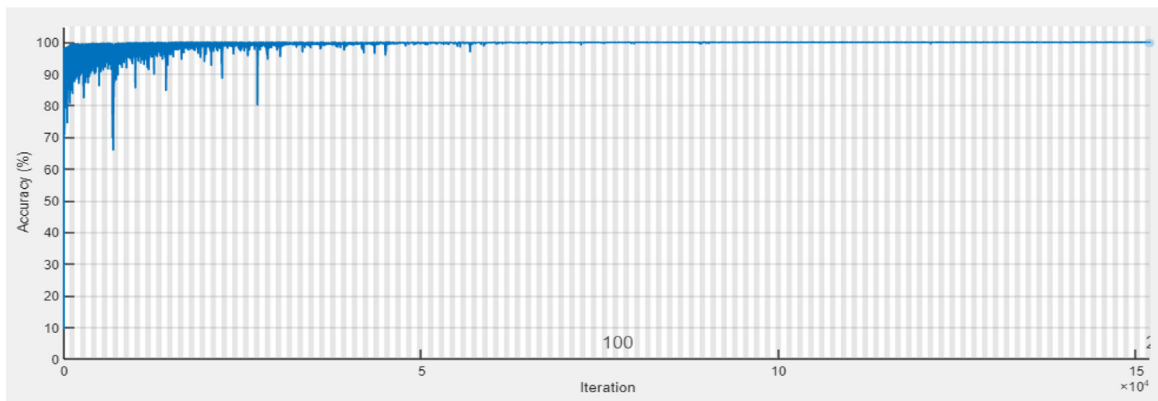


Fig. 3. Accuracy vs Iterations.



Fig. 4. Loss vs Iterations.

Table 1

Liver comparison of the segmentation on LiTS datasets.

Method	DSC (%)	ACCURACY (%)	VOE (%)	RVD (%)
Chen et al. [18]	95.7	–	–	–
Bi et al. [19]	93.4	–	–	–
Vorontsov et al. [20]	95.1	–	–	–
Yuan et al. [21]	96.3	–	–	–
Nanda et al. [22]	95.5	–	–	–
Chlebus et al. [23]	96.00	–	–	–
Omar Ibrahim Alirri [24]	95.6	99.31	–	–
Proposed method	96.35	99.71	6.89	6.30

4.2. Testing and evaluation

The developed method was trained on a machine with 8 GB RAM on an Intel Core i5-1035G1 CPU @ 1.00 GHz–1.19 GHz, and developed with MATLAB 2020a software. The images in the database were separated randomly into two groups for training and testing by the share 8.5:1.5. The following Table 1 shows the assessment metrics for the testing images. By utilising single CPU with a learning rate of 0.001 the network trained for 200 epochs where each epoch has taken 760 iterations. Training progress details which display accuracy vs iterations & loss vs iterations shown in Figs. 3 and 4. We observed as the number of training images increases the quality of segmentation also increases.

Liver segmentation results

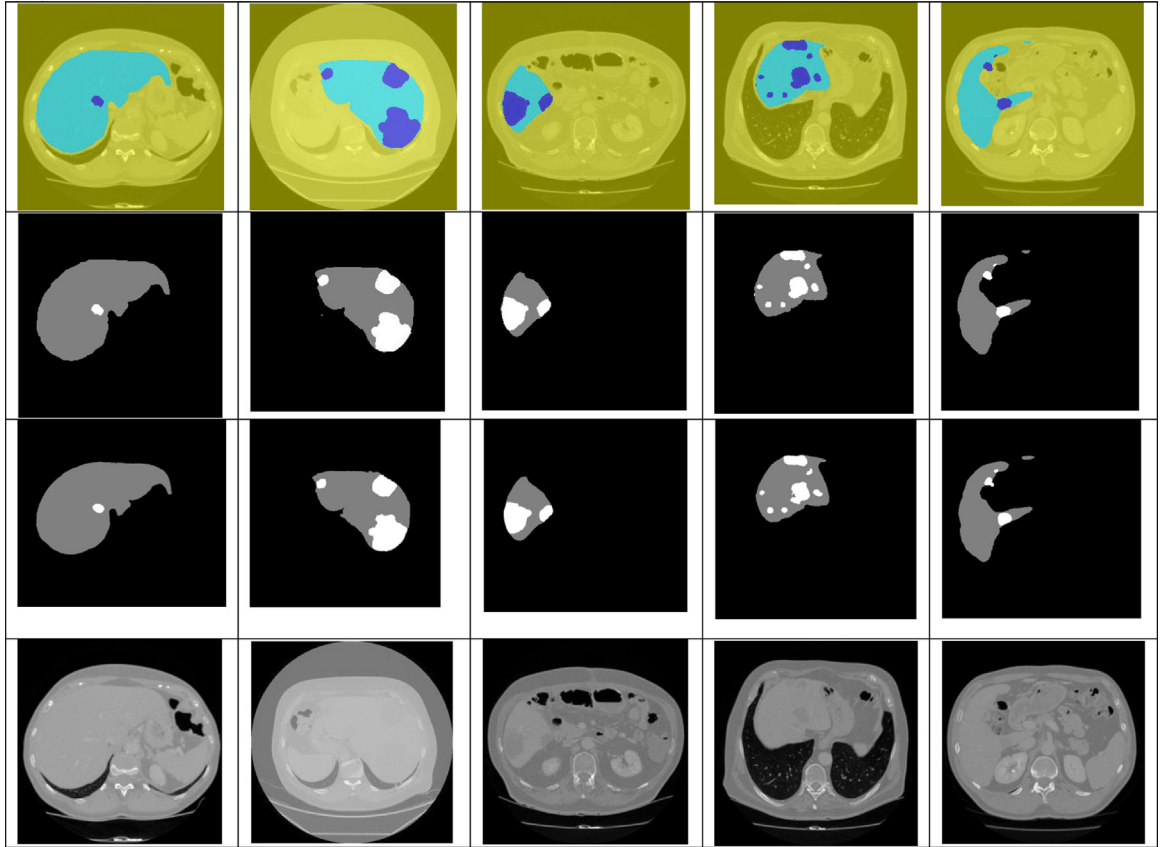
Different performance metrics are evaluated for the liver segmentation algorithm and the results are summarised above in Table 1. Result of the segmentation algorithm along with ground truths and input images are included in Fig. 5 where it has achieved a dice score of 96.35% and accuracy of 99.71%.

Here row 1 displays model results where liver is in light blue colour in the CT image, row 2 displays segmented liver result along with tumour, row 3 shows the original ground truths, and row 4 shows the original input images taken for testing.

Table 2

Liver tumour comparison of the segmentation on LiTS datasets.

Method	DSC (%)	ACCURACY (%)	VOE (%)	RVD (%)
Chen et al. [18]	66.6	–	–	–
Bi et al. [19]	64.5	–	–	–
Vorontsov et al. [20]	66.1	–	–	–
Yuan et al. [21]	65.7	–	–	–
Nanda et al. [22]	69.7	–	–	–
Chlebus et al. [23]	67.6	–	–	–
Omar Ibrahim Alirr. [24]	74.00	99.54	–	–
Proposed method	89.28	99.72	17.61	13.68

**Fig. 5.** Results of liver segmentation, ground truths and input image.

Tumour segmentation results

Different performance metrics are evaluated for the liver tumour segmentation algorithm and the results are summarised above in Table 2. Result of the segmentation algorithm along with tumour ground truths and liver input images are included in Fig. 6 where it has achieved a dice score of 89.28% and accuracy of 99.72%.

Here row 1 displays model results where liver tumour is in blue colour in the CT image, row 2 displays result of segmented liver tumour, row 3 shows the original ground truths of liver tumour, and row 4 shows the original liver input images taken for testing.

5. Conclusion

Liver & tumour segmentation algorithm based on deep learning is discussed in this paper. According to the fully convolutional network model, we construct a network called modified ResUNet to segment livers and its tumours. The main contributions of the paper are that the modified ResUNet outclasses other architectures in the occasion of liver extraction and tumour segmentation with a very high DICE score. In this work we used the modified ResUNet model for pixel-wise automatic liver and tumour segmentation using CT scans. CNNs helped us achieve our goals and are one of the best techniques that can be used in the segmentation of liver & its tumours where the ResUNet showed very promising results. By using more datasets and wide variety of pre-processing

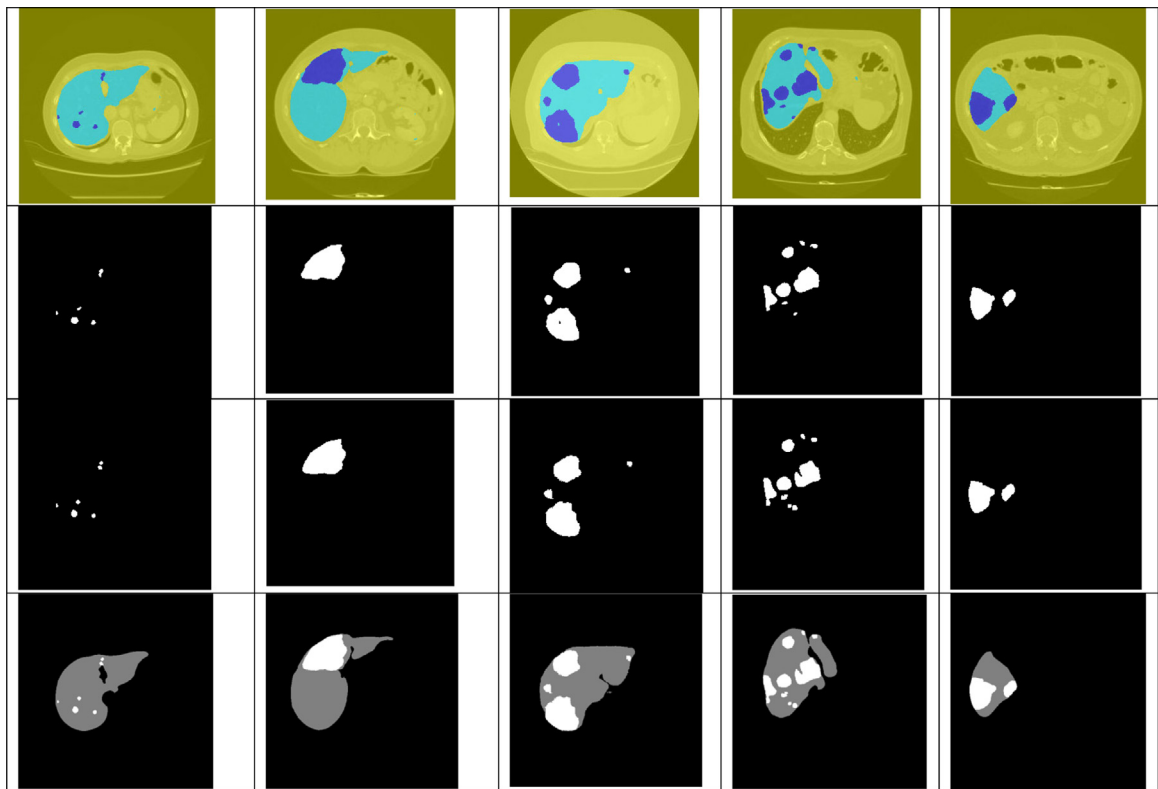


Fig. 6. Tumour segmentation results, ground truths and input image.

techniques, different parameters related to liver and its tumours can be improved using modified ResUNet model. In the future, different dataset will be investigated to check the system robustness and more CNN models will be tested to improve the performance and identification of smallest tumours.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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